Evaluating conformal prediction on compositional outputs

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Abstract

Outputs on the probability simplex arise as targets (proportions, compositions) and as predictions (classifier softmax vectors), and split conformal prediction offers them marginal coverage for any nonconformity score. It is less clear how reliably the *choice* of score, such as Euclidean, Hellinger, Aitchison or likelihood-based, can be evaluated, because conditional coverage must be estimated from finite data. We study this with synthetic Dirichlet and logistic-normal responses, for which the true conditional coverage and the oracle region are known, and with classifier sets on synthetic and CIFAR-10 outputs. Marginal coverage follows the exact Beta–binomial law for all 15 scores tested and carries no information on which score is better (rank correlation between replication blocks -0.03, against 0.94 for region volume). The worst-cell coverage estimator is biased low by an amount we give in closed form (mean -0.026 for ten cells and 500 test points), the bias differs between scores, and it can reorder scores (Kendall τ between the mean estimate and the true ranking 0.83–0.92); a single evaluation recovers the true ranking with Kendall $\tau \approx 0.5$–0.7. A half-split bias correction, tried because of this, overcorrects and does not improve the ranking. Normalised and likelihood scores give the smallest regions and flattest conditional coverage on Dirichlet data, whereas Aitchison-distance scores win on homoscedastic logistic-normal data and under-cover near the boundary. For classifier outputs the same small-cell bias appears; LAC sets are smaller than APS sets (paired interval above zero) in all 36 synthetic cells, and APS has the higher worst size-stratum coverage in 24 of them, the lower in 10. All results are for known-truth synthetic data, a fixed log-ratio linear mean model and one CNN; we state where they should not be extrapolated.

1 Introduction

Many outputs of machine-learning systems are compositions: vectors of non-negative parts that sum to one. Softmax class probabilities, topic proportions and species mixes are such vectors, as the quantity to be predicted or as the prediction itself. Conformal prediction [Vovk et al., 2005, Angelopoulos and Bates, 2023] turns any predictor into a set or region with finite-sample marginal coverage under exchangeability, and it is widely used for classifier outputs [Romano et al., 2020, Angelopoulos et al., 2021]. For a response that itself lies on the simplex, a practitioner must choose a nonconformity score, and the natural candidates come from different geometries: Euclidean and L_1 distances, the Hellinger distance, divergences, the Aitchison distance of log-ratio analysis [Aitchison, 1982], and the likelihood under a Dirichlet model.

Marginal coverage is guaranteed for all of them, so it cannot decide among them. What distinguishes scores is the size of the region and how evenly the coverage is spread over covariates and over the simplex, in particular near its boundary. Exact conditional coverage is impossible in a distribution-free

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setting [Vovk, 2012, Barber et al., 2021a], so papers report relaxations that are *estimated* from a finite test set: worst-slab coverage [Cauchois et al., 2020], size-stratified coverage [Angelopoulos et al., 2021], class-conditional gaps [Ding et al., 2023]. How reliable these estimates are is the question we ask, in settings where the true conditional coverage and the oracle region are known.

Research questions. **RQ1** Does the coverage of split conformal prediction follow its exact finite-sample law for every score on the simplex, and what does this imply for the sizes needed to certify coverage? **RQ2** At equal marginal coverage, how do score geometries differ in region size and in conditional and boundary coverage, and under which noise models is each preferable? **RQ3** How reliable are the estimators of worst-cell coverage used to compare scores: how large is their bias, does it depend on the method, and can it reverse a ranking? **RQ4** Under which conditions do the findings stop holding (covariate shift, rounding and ties, a misspecified mean, the calibration scheme)? **RQ5** Do the conclusions carry over to classifier outputs and their set-valued predictors?

Contributions. To our knowledge, and within the sources we consulted (Section 2), the following are new in combination.

- A known-truth benchmark for conformal regions on the simplex: 15 scores from Euclidean to Dirichlet-likelihood, two noise families, 20 configurations, with ground-truth conditional coverage and volume ratios to the oracle highest-density region (Section 4.2).
- An exact expression for the expected value of the worst-cell coverage estimate (Proposition 2), checked against simulation, and evidence on how the bias differs between methods, how often it reverses a pair of scores as a function of their true gap, and how many replications make a ranking reproducible (Section 4.3).
- Negative results reported as found: a half-split and a plug-in correction of the worst-cell estimator do not improve ranking, and a Mahalanobis score gives no gain over the Aitchison score.
- A transfer to classifier sets (LAC, APS, RAPS, Mondrian) including a CIFAR-10 CNN, an evaluation protocol, and conditions (covariate shift, rounding, a misspecified mean, CV+/jackknife+) where results change (Sections 4.4–5).

The exact coverage law (Proposition 1) is a known property of split conformal prediction; we use it as the reference.

Scope. Data are synthetic with a known truth, except one CIFAR-10 network; we do not claim that rankings carry over to real compositional data.

2 Related work

Conformal prediction. Conformal prediction [Vovk et al., 2005] gives finite-sample marginal coverage under exchangeability; the inductive (split) variant [Papadopoulos et al., 2002, Lei et al., 2018] separates fitting from calibration, and cross-conformal methods such as the jackknife+ [Barber et al., 2021b] trade computation for sample efficiency. Weighted conformal prediction handles known covariate shift [Tibshirani et al., 2019].

Set-valued classification. For classifier outputs $\hat{p}(x) \in \Delta^{K-1}$, the least-ambiguous-set score [Sadinle et al., 2019], adaptive prediction sets [Romano et al., 2020] and regularised variants [Angelopoulos et al., 2021] differ in conditional behaviour, not in marginal validity. Class-conditional and clustered calibration targets conditional coverage under many classes [Ding et al., 2023], and multivalid methods target coverage over rich groups [Jung et al., 2022]. Credal-set predictors return sets of distributions [Javanmardi et al., 2024].

Conditional coverage: limits and how it is measured. Exact distribution-free conditional coverage is impossible for non-atomic features, and approximate versions require relaxations [Vovk, 2012, Barber et al., 2021a]; coverage conditional on the training set is a separate guarantee [Bian and Barber, 2023]. Practice therefore reports relaxed, *estimated* summaries: the minimum coverage over feature slabs (worst-slab coverage, Cauchois et al., 2020), the maximal deviation of coverage across set-size strata [Angelopoulos et al., 2021], and the class-conditional coverage gap [Ding et al., 2023].

Braun et al. [2025] note in an appendix that the worst-slab estimate is systematically below the target even under exact conditional coverage, since the expectation of a minimum is at most the minimum of expectations, and propose classifier-based diagnostics. In contrast, Cauchois et al. [2020] report in one multilabel simulation that the worst-slab ordering of methods agreed with the ordering by exact conditional coverage. We take the first as established (Proposition 2 makes it quantitative) and test the second: whether a method-dependent bias can change a ranking.

Compositional and multivariate responses. Compositional data analysis treats observations on the simplex through log-ratio geometry [Aitchison, 1982, 1986]; zeros require special handling, and we use the geometries as score choices. For vector-valued responses, conformal regions have been built from ellipsoidal scores based on residual covariances [Henderson et al., 2024], from optimal-transport ranks of multivariate scores [Klein et al., 2025], and, for outputs linked by linear constraints, from a projection step that provably shrinks split-conformal regions [Principato et al., 2024]. These works target $\mathbb{R}^d$ or linear structure; the non-negativity boundary and the choice of geometry are not their subject as far as the abstracts we read go. Our search was limited and deliberately excluded work on conformal prediction with compositional responses (Appendix B); we make no claim about its absence.

Gap. To our knowledge and within the sources consulted, three points are not addressed together: classification-set evaluation is defined for a discrete label, while here the response is continuous, so set size becomes a volume relative to an oracle region; the bias of the worst-cell estimator, and whether it can reverse a ranking, has not been quantified where the true conditional coverage is known; and the choice of geometry for a simplex-valued response is usually left to the analyst, although rank-based coverage is invariant to monotone score transformations but not to a change of geometry.

3 Setting, scores and the evaluation problem

Compositional outputs. We call an output *compositional* if it lies in the simplex $\Delta^{K-1} = \{p \in \mathbb{R}^K_{\geq 0} : \sum_k p_k = 1\}$. Two settings are studied. In Setting A the *target* is a composition, $Y \in \Delta^{K-1}$, and a method returns a region $C(X) \subset \Delta^{K-1}$. In Setting B the *prediction* is a composition: a classifier returns $\hat{p}(x) \in \Delta^{K-1}$, the target is a label $Y \in \{1, \ldots, K\}$, and a method returns a label set $C(x) \subset \{1, \ldots, K\}$.

Split conformal prediction. Let a nonconformity score $s(x, y)$ be fitted on a training split, let $S_1, \ldots, S_n$ be the scores on a calibration split of size n, and let S_{n+1} be the score of a test pair. For miscoverage level α set $k = \lceil (n+1)(1-\alpha) \rceil$ and $\ell = n + 1 - k = \lfloor (n+1)\alpha \rfloor$. The threshold is $\hat{q} = S_{(k)}$ (the k-th smallest calibration score; $\hat{q} = \infty$ if $k > n$) and $C(x) = \{y : s(x, y) \leq \hat{q}\}$.

3.1 The exact law of coverage

Define the conditional coverage given the calibration data $V = \mathbb{P}\{S_{n+1} \leq \hat{q} \mid S_1, \ldots, S_n\}$.

Proposition 1 (coverage law)**.** *If $S_1, \ldots, S_{n+1}$ are exchangeable with a continuous joint law (no ties almost surely), then $V \sim \mathrm{Beta}(n + 1 - \ell,\ \ell)$. Consequently (i) $\mathbb{E}V = (n + 1 - \ell)/(n + 1) \in [1 - \alpha,\ 1 - \alpha + \frac{1}{n+1})$; (ii) $\mathrm{Var}\, V = \dfrac{(n + 1 - \ell)\,\ell}{(n+1)^2(n+2)} \approx \dfrac{\alpha(1-\alpha)}{n+2}$; and (iii) the number of covered points among m fresh test points, given the calibration data, is $\mathrm{Bin}(m, V)$, so unconditionally it is $\mathrm{BetaBinomial}(m,\ n + 1 - \ell,\ \ell)$. This law is the same for every score function s, hence for every choice of geometry on Δ^{K-1}.*

The law is a known property of split conformal prediction [Vovk, 2012, Angelopoulos and Bates, 2023]; we state it because the rest of the paper uses it as the reference against which every score, geometry and estimator is checked.

Proof. For i.i.d. scores with continuous cdf F, $U_i = F(S_i)$ are i.i.d. uniform and $V = F(S_{(k)}) = U_{(k)}$ is the k-th order statistic of n uniforms, which is $\mathrm{Beta}(k, n + 1 - k) = \mathrm{Beta}(n + 1 - \ell, \ell)$. For merely exchangeable scores with no ties, the same holds conditionally on the order statistics by the usual randomisation argument (the rank of S_{n+1} among the $n + 1$ scores is uniform), and V depends

only on ranks. The mean and variance are those of the Beta law, and (iii) follows because test points are, given the calibration data, i.i.d. Bernoulli(V) for i.i.d. data. □

Remark 1 (ties)**.** *If the score has atoms (rounded compositions, zeros replaced by a constant, discrete $\hat{p}$), coverage remains $\geq 1-\alpha$ but the upper bound in (i) is lost, and coverage is conservative by a score-dependent amount (tested in Section 4).*

What this implies for evaluation. (a) Marginal coverage cannot separate score geometries, which differ only through region size and conditional coverage. (b) The spread of the *measured* coverage is dominated by the calibration draw, $\mathrm{sd} \approx \sqrt{\alpha(1-\alpha)/(n+2)}$, which a larger test set does not remove. (c) Certifying $|V-(1-\alpha)| \leq \varepsilon$ with probability $1-\delta$ requires the n with $\mathbb{P}(|V-(1-\alpha)| \leq \varepsilon) \geq 1-\delta$, computed from the Beta cdf; the normal approximation gives $n \gtrsim \alpha(1-\alpha) z^2_{1-\delta/2}/\varepsilon^2$.

3.2 Bias of worst-cell coverage estimators

Conditional coverage is evaluated by partitioning the feature space (or the set-size range, or the label set) into G cells and reporting the worst cell. Condition on the fitted model and calibration data, let c_g be the true coverage of cell g, and let m_g test points fall in it. The cell estimates are $\hat{c}_g = B_g/m_g$ with $B_g \sim \mathrm{Bin}(m_g, c_g)$ independent across cells (disjoint test points).

Proposition 2 (worst-cell estimator)**.** *For equal cell sizes $m_g = m$,*

$$\mathbb{E}\Big[\min_g \hat{c}_g\Big] = \frac{1}{m}\sum_{j=1}^{m}\prod_{g=1}^{G}\mathbb{P}\{\mathrm{Bin}(m, c_g) \geq j\},$$

which is at most $\min_g c_g$, with strict inequality whenever some $\hat{c}_g$ is non-degenerate. If all $c_g = c$ and m is large, $\mathbb{E}[\min_g \hat{c}_g] \approx c - \sqrt{c(1-c)/m}\ \mu_G$ with $\mu_G = \mathbb{E}\max_{g\leq G} Z_g$ for i.i.d. standard normals ($\mu_{10} \approx 1.54$).

Proof. $\min_g \hat{c}_g$ takes values in $\{0, 1/m, \ldots, 1\}$, so $\mathbb{E}\min = \sum_{j=1}^{m} \frac{1}{m}\mathbb{P}\{\min_g B_g \geq j\}$ and independence gives the product. The inequality is $\mathbb{E}\min_g \hat{c}_g \leq \min_g \mathbb{E}\hat{c}_g = \min_g c_g$ (Jensen for the concave min). The Gaussian approximation is the expected maximum of G i.i.d. normals applied to $-(\hat{c}_g - c)/\sqrt{c(1-c)/m}$. □

The bias grows with G and shrinks with $m = n_{\mathrm{te}}/G$ and is of the order of the differences between methods one hopes to resolve; rankings are affected through differences in the spread of c_g across methods, not through a common offset.

4 Experiments

4.1 Synthetic simplex targets and the exact coverage law

Design. Covariates are $X \sim N(0, I_4)$ and the conditional mean is $\mu(x) = \mathrm{softmax}(Wx + b)$ with fixed population coefficients. We use $K = 3$ and two noise models with known conditional laws: Dirichlet, $Y \mid x \sim \mathrm{Dir}(\phi(x)\mu(x))$, and logistic-normal, with isotropic Gaussian noise in an orthonormal log-ratio basis; in both the scale varies with x_0 (heteroscedasticity strength $h = 0.5$). Every replication draws fresh training (500), calibration (n) and test (2000) samples, fits the mean by ordinary least squares in clr coordinates, and applies split conformal prediction at $\alpha = 0.1$ with 15 scores on the same draws: Euclidean, L_1, Hellinger, Aitchison (clr) and forward and reverse KL, each raw and normalised by a fitted local scale, plus quantile-normalised Euclidean and Aitchison and a Dirichlet likelihood score. We run $R = 2000$ replications per setting and calibration size $n \in \{20, 50, 100, 200, 500, 2000\}$.

Coverage follows the exact law for every score. For each replication the observed test coverage is mapped through the randomised probability integral transform under the Beta-binomial law of Proposition 1; under the law the transformed values are uniform. Figure 4 (Appendix A.1) shows the empirical cdf minus the identity for all 15 scores, three calibration sizes and both noise models. Across the six panels shown the curves of all scores are close to the zero line and mostly inside the

95% DKW band of a single ecdf for $R = 2000$. Over all 12 settings and 15 scores (180 Kolmogorov–Smirnov tests, Table 2 in Appendix A.1), 2 rejected at the 5% level (logistic-normal noise at $n = 100$ and $n = 200$, $p = 0.047$ and 0.045); the tests are dependent because the scores share draws, so the count has no binomial reference, but it is below the 9 that independent tests of a true law would give. The ratio of observed to exact standard deviation of coverage is within $[0.988, 1.014]$ in every row of the table (medians over the 15 scores). The calibration-draw spread is therefore the predicted one, whatever the region size.

How large must a calibration set be to certify coverage? By Proposition 1, the smallest n with $\mathbb{P}(|V - (1-\alpha)| \le \varepsilon) \ge 0.95$ is 3448 for $\alpha = 0.1, \varepsilon = 0.01$ and 854 for $\varepsilon = 0.02$, against 3457 and 864 from the normal approximation (`certify_n.csv`; simulation gives 0.9504 at $n = 854$); the approximation is poor at $\varepsilon = 0.05$, $\alpha = 0.05$ (29 exact vs. 73).

4.2 Does the geometry of the score matter? (RQ2)

Design. Experiment A1 varies one factor at a time around a base configuration (Dirichlet noise, $K = 3$, $h = 0.5$, precision 20; 500 training, 500 calibration, 10^4 test points, $\alpha = 0.1$): noise family, $K \in \{3, 5, 10\}$, $h \in \{0, 0.5, 1\}$, linear or quadratic scale, precision or σ, and two-digit rounding with zeros replaced by $\delta = 10^{-4}$; 20 configurations, 100 or 200 replications each. Because the conditional law is known, the quantities that finite data cannot give are computed on a fixed grid of 3000 covariate values (300 per decile of x_0, 30 draws of y each): the true coverage of each slab and of ten boundary strata (deciles of the smallest predicted component), and the $\log_{10}$ volume ratio to the oracle highest-density region of the same level (only for $K \le 5$). The worst true slab coverage of the best scores is only 0.86–0.89 because the calibration set is finite, which sets the scale of the differences below.

Marginal coverage does not separate the scores; size and conditional coverage do. Across the 20 configurations and 15 scores every estimated marginal coverage lies within 0.0035 of 0.9 (within 0.0059 on the ground-truth grid). Ranking scores by their marginal-coverage gap is not reproducible: the Kendall τ between rankings from two disjoint blocks of $R = 25$ replications is -0.03 on average, against 0.94 for the volume ratio and 0.89 for the true worst-slab coverage (`A3_rank_blocks.csv`). At the same marginal coverage the regions differ widely (Fig. 1a,b; Appendix A.2 for all h): in the base configuration the volume is $10^{0.02}$ of the oracle's for the Dirichlet score, $10^{0.12}$ for Hellinger, $10^{0.21}$ for Euclidean and $10^{0.42}$ for Aitchison, and the worst true slab coverage ranges from 0.63 (Aitchison) to 0.84 (Dirichlet score).

Normalised and likelihood-based scores come closest to the oracle region. On Dirichlet data the Dirichlet score has a volume ratio of $10^{0.02}$–$10^{0.17}$ against $10^{0.18}$–$10^{0.34}$ for Euclidean, with a paired 95% interval excluding zero in 17 of the 18 configurations with a computed volume; its gain in worst-slab coverage is positive in 18 of 20 configurations but modest (mean 0.07, 0.78 vs. 0.75 at $h = 1$). Dividing a score by a fitted local scale mainly repairs conditional coverage: for logistic-normal data at $h = 1$ the worst slab rises from 0.53 (Euclidean) to 0.80 (Euclidean$/n$) and 0.81 (quantile-normalised), while the volume ratio falls from $10^{0.78}$ to $10^{0.52}$ and, for the normalised Aitchison score, to $10^{0.30}$. When the likelihood family is wrong the Dirichlet score keeps small regions on logistic-normal data (volume $10^{0.09}$–$10^{0.53}$) but loses conditional coverage when heteroscedastic (worst slab 0.65–0.78 against 0.84–0.87 for the quantile-normalised Aitchison score).

The best geometry depends on the noise model and on the boundary. No score is best everywhere (Fig. 1c,d; Table 3 in Appendix A.6). Aitchison scores give a *larger* region than Euclidean in 13 of the 18 computed configurations and a smaller one in 5; on homoscedastic logistic-normal data, whose noise is isotropic in clr coordinates, they are the best ($10^{0.08}$ vs. $10^{0.31}$, $K = 3$). On Dirichlet data their worst boundary stratum is covered at only 0.37–0.61 (nine unrounded configurations), against at least 0.76 for Euclidean, L_1, Hellinger, the two KL scores, the Dirichlet score and the normalised Euclidean and Hellinger scores; with two-digit rounding it is 0.81 and depends on the zero replacement: from 0.59 at $\delta = 10^{-6}$ to 0.86 at $\delta = 10^{-2}$ on logistic-normal data (0.80 to 0.86 on Dirichlet data), whereas Euclidean, L_1 and Hellinger move by at most 0.02 (Fig. 7 in Appendix A.4). Hellinger and forward-KL give smaller regions than Euclidean in all 18 configurations. A Mahalanobis score on the clr residual covariance adds nothing here (within 0.07, 0.03, 0.09 of Aitchison

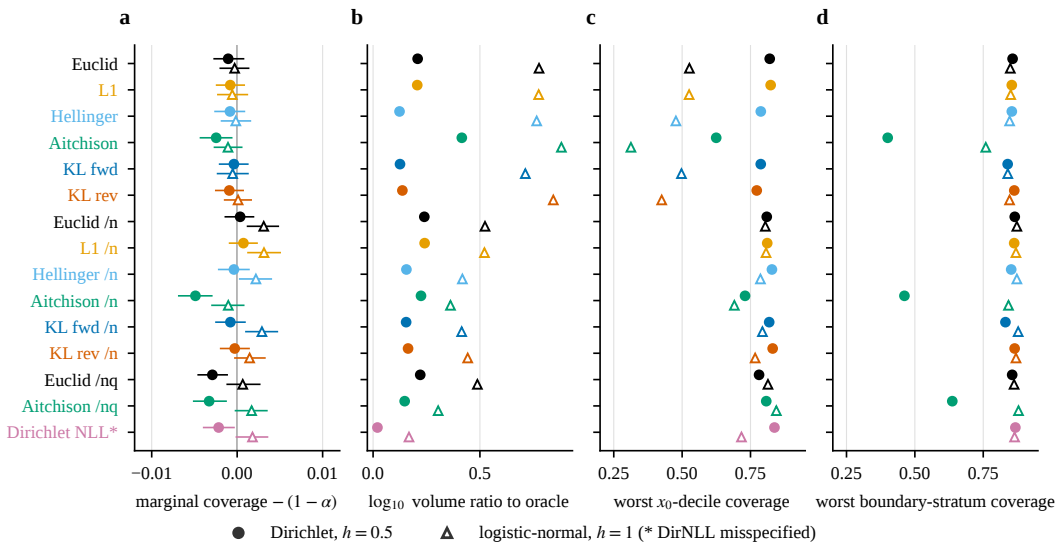


Figure 1: Marginal coverage is flat across scores; region size and conditional coverage are not. One row per score (colour = geometry; /n and /nq: normalised, quantile-normalised), two A1 configurations with $K = 3$: Dirichlet, $h = 0.5$ (filled) and logistic-normal, $h = 1$ (open; the Dirichlet-NLL score is misspecified there). (a) True marginal coverage minus 0.9. (b) $\log_{10}$ volume ratio to the oracle region. (c,d) True coverage of the worst x_0-decile and of the worst boundary stratum. Whiskers: 95% bootstrap intervals over replications. Source: `figures/fig_c3c4_main.py`.

in volume, worst slab, worst boundary stratum; `A1m_summary.csv`), which is expected for isotropic noise.

4.3 How reliable are the estimators used to compare scores? (RQ3)

The worst-cell estimate is biased low, and Proposition 2 predicts by how much. Estimating the worst of G cell coverages from a test set of n_{te} points is biased downwards whatever the score. In A1 the mean bias of the worst-slab estimate ($G = 10$, averaged over 20 configurations and 15 scores) is -0.026, -0.005 and $+0.002$ at $n_{\text{te}} = 500$, 2000 and 10^4; for the worst boundary stratum it is -0.041, -0.014, -0.001, and for $G = 20$ slabs -0.063, -0.017, 0.000 (`A3_bias.csv`). Evaluating the exact formula of Proposition 2 with each replication's true cell coverages reproduces the observed mean estimate with a mean absolute error of 0.005 (maximum 0.045; `A3_pred_bias.csv`). The Gaussian approximation is accurate only for equal cell coverages: with equal c_g its error is at most 0.012 at $m = 25$ points per cell and $G = 20$, with unequal coverages it overestimates the expected minimum by 0.025–0.09 (`prop2_equal_cells.csv`); we therefore use the exact formula (Fig. 8 in Appendix A.5).

The bias depends on the score and can reorder scores. Across the 15 scores the biases of the worst-slab estimate span 0.063, 0.037 and 0.029 at $n_{\text{te}} = 500$, 2000, 10^4, while the true worst-slab coverages span 0.29 (Fig. 2a). With the sampling noise averaged out, the Kendall τ between the mean estimate and the true ranking is 0.83, 0.88, 0.92 (the ranking predicted from the exact formula: 0.91, 0.95, 0.98): mis-ranking is real. How often it reverses a pair falls with the true gap (Appendix A.9; worst-decile coverage, 105 score pairs, 8 configurations, 1000 replications). For a single replication at $n_{\text{te}} = 500$, 2000, 10^4, a pair 0.03–0.05 apart is reversed with probability 0.26, 0.10, 0.02 and a pair 0.10–0.20 apart with 0.05, 0.002, 0; reversals stay below 5% only for gaps ≥ 0.2, 0.05, 0.03. For the mean of 25 replications a pair 0.03–0.05 apart is reversed with probability 0.03, 0.005, 0.001. A single evaluation at $n_{\text{te}} = 500$ thus reverses pairs up to about 0.2 apart with probability $\geq 5\%$; with 25 averaged replications or $n_{\text{te}} \geq 2000$ only pairs closer than about 0.01–0.03 are reversed (`A3_flip_threshold.csv`).

Table 1: Protocol for comparing conditional coverage of scores, with the supporting numbers of this section (block τ: mean over configurations, as in the text). Source: `tables/tab_c8_protocol.py`.

item	recommendation	supporting number
Marginal coverage	report it as a validity check only; do not use it to compare scores	Kendall τ between replication blocks (mean over configs) $= -0.03$ ($R = 25$), vs 0.94 for volume
Calibration size	choose n from the exact Beta law; $n = 854$ certifies $\lvert\text{cov} - 0.9\rvert \le 0.02$ w.p. 0.95	`certify_n.csv`, $\alpha = 0.1$
Conditional coverage	report cell size $m = n_{te}/G$ and the bias of Proposition 2 for assumed cell coverages (plug-in version untested); require $m \ge 200$	naive worst-cell bias, $G = 10$: -0.026 ($m = 50$), -0.005 ($m = 200$), $+0.002$ ($m = 1000$); $G = 20$, $m = 100$: -0.017
Replications	at least 25 independent replications; rank by paired differences	τ between blocks at $R = 5/10/25$: $0.78/0.84/0.89$ (true worst decile, mean over 20 configs)
Pairing	evaluate all scores on the same draws; use paired bootstrap differences	median SE ratio unpaired/paired: 1.51 (volume), 1.29 (worst decile)
Do not correct by splitting	naive estimate, not a half-split estimator	bias of true minimum, $G = 10$, $n_{te} = 2000$: naive -0.002, half-split $+0.013$
Resolution	state which contrasts are unresolved at the chosen R	113 of 1170 score-vs-Euclid contrasts (10%) have a paired 95% CI containing 0 at $R = 100$–200

Single evaluations rank poorly; replications and paired designs are what matter. Ranking the 15 scores by the worst-slab estimate from *one* replication agrees with the true ranking with $\tau = 0.50, 0.60, 0.66$ and a correct top score in 23%, 31%, 36% of replications at $n_{\text{te}} = 500$, 2000, 10^4 (`A3_rank_single.csv`). Averaging R replications helps: the τ between two disjoint blocks reaches 0.81 at $n_{\text{te}} = 2000$ and 0.87 at 10^4 for $R = 25$ (Fig. 2b). Pairing tightens the comparison (median unpaired/paired standard error 1.51 for volume, 1.29 for the worst slab; Fig. 2c), yet 113 of the 1170 contrasts with Euclidean in A1 (10%) still have a paired interval containing zero.

A natural bias correction fails. We tried selecting the apparent worst cell on one half of the test set and measuring it on the other ("half-split", both directions). It is unbiased for the coverage of the cell it selects (bias ≤ 0.004), but it is biased *upwards* for the true minimum ($+0.029$, $+0.013$, $+0.007$ at $G = 10$ and $n_{\text{te}} = 500$, 2000, 10^4, against -0.018, -0.002, $+0.003$ for the naive estimate), and it does not improve the ranking: τ for a single replication is 0.56 against 0.62 for the naive estimate at $n_{\text{te}} = 500$ and 0.75 against 0.76 at 10^4 (8 configurations, 1000 replications; Fig. 2d). Evaluating Proposition 2 at cell coverages estimated on the same test set (plug-in) also does not help: it removes the mean bias (residual $+0.0003$ against -0.018, worst decile, $n_{\text{te}} = 500$) but not the error (RMSE 0.0596 against 0.0601) or the ranking (single-replication τ 0.60 against 0.62, mean over 8 configurations), and it predicts only 0.027 of the 0.070 spread of the bias across scores (Appendix A.10). We therefore recommend reporting the cell size and the size of the bias, not subtracting it.

Table 1 collects what these results imply for evaluating conditional coverage on simplex data.

4.4 Do the findings carry over to classifier outputs? (RQ5)

A softmax vector is a point of the simplex Δ^{K-1}, and the sets LAC, APS, RAPS and class-conditional (Mondrian) LAC are split-conformal regions for the label given the vector. We use two sources. *B-synth*: $K = 10$, true label probabilities $p \sim \text{Dir}(\beta\mathbf{1})$ with $\beta \in \{0.1, 0.3, 1\}$, classifier output $\hat{p} = \text{softmax}(\tau \log p + \varsigma\varepsilon)$ with $\tau \in \{0.5, 1, 2\}$ and $\varsigma \in \{0, 1\}$, calibration size $n \in \{100, 1000\}$ (36 cells, 500 replications), so that the exact coverage given $\hat{p}$ is known. *B-cifar*: a small CNN trained on 40 000 CIFAR-10 images, with 20 000 held-out images (accuracy 0.869) with temperature $T \in \{0.5, 1, 2\}$, $n \in \{100, 1000\}$ and 1000 random splits of that one network (the spread is split variance, not training variance). The conditional-coverage measure is the minimum coverage over strata of the set size (sizes 1, 2, 3–4, 5–10); results are in Fig. 3.

LAC gives the smallest sets, but not always the worst size-stratified coverage. LAC sets are smaller on average than those of APS, RAPS and Mondrian-LAC in all 36 B-synth cells (for example 6.04 vs. 6.18 labels at $\beta = 1$, $\tau = 0.5$, $\varsigma = 0$, $n = 1000$); the paired 95% interval of the size difference lies above zero in all 36 cells for APS and Mondrian-LAC and in 34 for RAPS

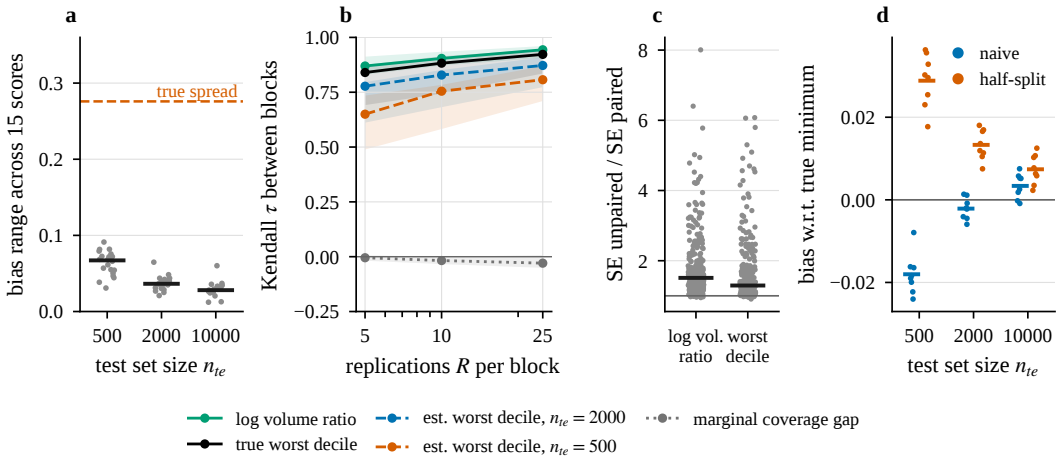


Figure 2: Rankings of scores need replications and pairing, and the half-split correction does not help. (a) Range over the 15 scores of the bias of the worst-slab estimate, against the true spread of worst-slab coverage (dashed). (b) Kendall τ between rankings from two disjoint blocks of $R \leq 25$ replications (20 configurations), for the marginal-coverage gap, the log volume ratio, the true worst decile and the estimated worst decile ($n_{\text{te}} = 500, 2000$). (c) Unpaired over paired standard error of the contrast with Euclidean. (d) Bias of the naive and half-split estimate of the true minimum, $G = 10$, 8 configurations.

(`B_paired_vs_lac.csv`). The expected order of conditional coverage holds in most cells: the exact worst size-stratum coverage of APS is higher than that of LAC in 24 of 36 cells (paired interval above zero), lower in 10 and unresolved in 2. The exceptions are sometimes large: at $\beta = 0.3$, $\tau = 0.5$, $\varsigma = 0$, $n = 1000$ the worst stratum has coverage 0.865 for LAC and 0.805 for APS; at $\beta = 1$, $\tau = 0.5$, $\varsigma = 1$ it is 0.30 for LAC, 0.59 for APS and 0.90 for RAPS. On CIFAR-10 ($T = 1$, $n = 1000$) LAC has size 1.089 and worst size-stratum coverage 0.872 (largest class-conditional gap 0.104), APS 1.396, 0.911 and 0.024; RAPS equals APS within 0.003 in size and 0.002 in coverage here. Mondrian-LAC overcovers and inflates the sets at $n = 100$ (size 4.2 against 1.1, coverage 0.944).

Small strata bias the estimates of classifier sets in the same way. With $n_{\text{te}} = 200, 1000, 5000$ test points the estimated worst size-stratified coverage of CIFAR-10 is 0.066, 0.034, 0.009 too low for APS and 0.031, 0.015, 0.003 for LAC, measured against the same quantity on the whole pool (`B_ssc_bias.csv`). Restricting the minimum to strata with at least 30 test points removes most of it (mean bias relative to the same strata at most 0.011 in absolute value; the minimum cell size of Table 1).

4.5 Conditions: covariate shift, rounding, misspecified mean, calibration scheme

Covariate shift. Proposition 1 assumes exchangeability. When x_0 is shifted by δ standard deviations, unweighted split intervals under-cover (A4; known density ratio, four scores): at $\delta = 1$ coverage is 0.72–0.82 on Dirichlet and 0.66–0.88 on logistic-normal data, lowest for the Aitchison score, and density-ratio weighting restores 0.900–0.908. At $\delta = 1.5$ (effective sample size about 75) weighting overcorrects, to 0.904–0.925, because 2.8% of test points receive an infinite threshold (`A4_summary.csv`; Appendix A.8).

Rounding and ties. Rounding to one to three digits changes the mean coverage by at most 0.0014 with a fitted mean; true ties (intercept-only mean) raise it by 0.003–0.005 at one digit: conservative and small, as the remark after Proposition 1 says (`A2_beta_law.csv`; Appendix A.7).

Misspecified mean. With a quadratic part in the true logit mean (30% and 63% of its variance unexplained by the linear fit; $K = 3$; A6), marginal coverage stays within 0.891–0.901 for all 15 scores, but regions are 0.31–0.81 larger in $\log_{10}$ volume and the ranking of scores is not stable:

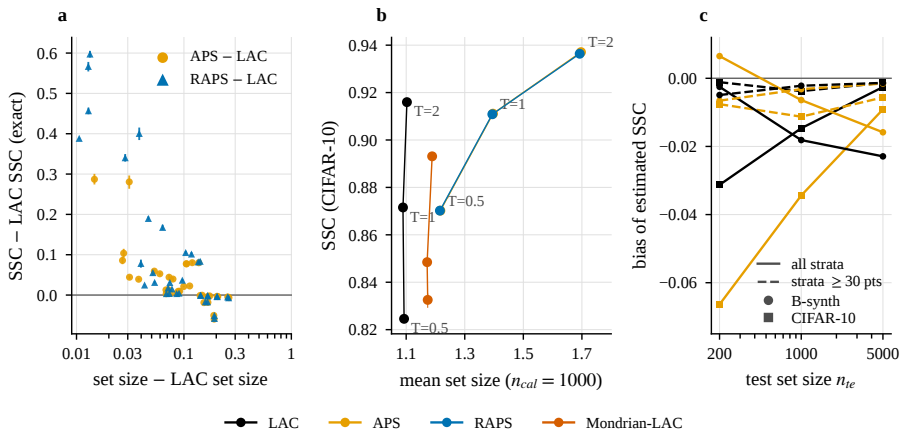


Figure 3: Classifier sets. (a) B-synth, 36 cells: difference in exact worst size-stratum coverage against difference in mean set size, APS and RAPS relative to LAC, with paired bootstrap intervals. (b) CIFAR-10, $n_{\text{cal}} = 1000$, mean set size against worst size-stratum coverage for $T \in \{0.5, 1, 2\}$; intervals: splits of one network. (c) Bias of the estimated worst size-stratified coverage against the test size, mean over configurations (B-synth, 36; CIFAR-10, 6), over all strata and over strata with at least 30 test points.

Kendall τ with the ranking under the correct mean is 0.07–0.39 by volume and -0.12 to 0.52 by worst-decile coverage (Appendix A.12). Which score is best is therefore not robust to a misspecified mean.

CV+ and the jackknife+. Against split calibration at $n = 30$ both cover less in point estimate in all 8 cells (interval below zero in 12 of 16 method–cell contrasts; -0.006 to -0.047; mostly because split overcovers, e.g. 0.94) and give smaller regions ($\log_{10}$ volume ratio -0.13 to -0.27); the gain shrinks to -0.01 to -0.03 at $n = 200$ (`A5_paired_vs_split.csv`; two noise models, four scores, 100 replications; jackknife+ not run at $n = 200$; Appendix A.11).

5 Discussion and limitations

Which method for which condition. Marginal validity does not depend on the score, so the choice should rest on region size and conditional coverage, evaluated with the protocol of Table 1. For Dirichlet-like noise with covariate-dependent scale, a normalised or likelihood score gives the smallest regions and the best worst-slab coverage; for homoscedastic noise isotropic in log-ratio coordinates the Aitchison distance is best, but it under-covers boundary strata and depends on the zero-replacement constant. A misspecified mean can reverse these rankings. For classifier sets, LAC gives the smallest sets and APS or RAPS the better worst size-stratum coverage in two thirds of our synthetic cells, with sizeable exceptions.

What the evaluation results mean for practice. A worst-slab coverage from one split is a biased and noisy summary: report the cell size and the size of the bias (Proposition 2), use cells of at least about 200 test points, and rank on at least 25 replications with paired differences (Table 1). The method-dependent bias reverses pairs of scores up to about 0.2 apart in true worst-decile coverage with probability $\geq 5\%$ in a single evaluation with 500 test points; with 25 averaged replications or $n_{\text{te}} \geq 2000$ only pairs closer than about 0.01–0.03.

Limitations and where not to extrapolate.

- *Synthetic known-truth data.* Two noise families, isotropic in log-ratio coordinates; real compositional data were not studied; classifier sets use one CNN (intervals reflect split variance only).
- *Mean model and dimension.* Linear clr regression; one misspecification (quadratic, $K = 3$) was probed; volume ratios exist for $K \leq 5$ only.

- *Other.* CV+ and the jackknife+: two noise models, four scores; counts of "better" and "worse" paired intervals are descriptive (no multiplicity correction); covariate shift: known density ratio, one dimension; limited literature search (Appendix B).

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A Additional results

A.1 The coverage law: ecdf and Kolmogorov–Smirnov tests

Observed coverage was tested against the Beta-binomial law of Proposition 1 for every score and noise model (Section 4, first paragraph).

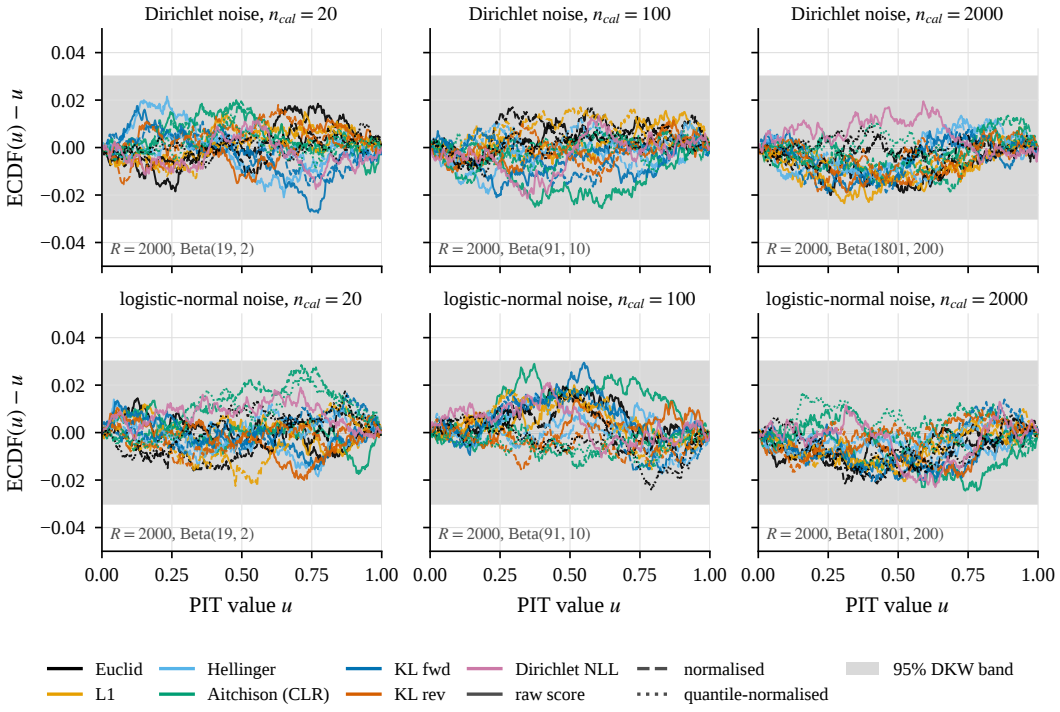


Figure 4: Observed split-conformal coverage follows the exact law for every score. Empirical cdf of the randomised PIT of the observed test coverage under $\mathrm{BetaBinomial}(2000, n+1-\ell, \ell)$, minus the identity (zero line = exact law); one curve per score (colour = geometry, line style = raw / normalised / quantile-normalised), $R = 2000$ replications, $\alpha = 0.1$. Grey: 95% DKW band for a single ecdf. Source: `figures/fig_c1_coverage_law.py`.

Table 2: Test of the exact coverage law. For each noise model and calibration size, the 15 scores are tested (randomised-PIT KS test, $R = 2000$, $\alpha = 0.1$, 2000 test points). The 15 tests share draws and are dependent. z: mean coverage minus the exact mean, in standard errors (median over scores); SD ratio: observed over exact standard deviation of coverage (median). Source: `tables/tab_c1_ks.py`.

noise	n_{cal}	R	$\#\{p < 0.05\}/15$	min p	median z	median SD ratio
Dirichlet	20	2000	0	0.088	+0.06	0.993
Dirichlet	50	2000	0	0.082	-0.59	1.014
Dirichlet	100	2000	0	0.110	+0.09	0.996
Dirichlet	200	2000	0	0.051	-0.70	1.006
Dirichlet	500	2000	0	0.095	-0.01	0.994
Dirichlet	2000	2000	0	0.193	+0.49	0.990
logistic-normal	20	2000	0	0.074	-0.35	1.011
logistic-normal	50	2000	0	0.072	+1.23	1.003
logistic-normal	100	2000	1	0.047	-0.38	1.012
logistic-normal	200	2000	1	0.045	-0.66	0.988
logistic-normal	500	2000	0	0.208	-0.65	1.007
logistic-normal	2000	2000	0	0.118	+0.67	0.988

A.2 Marginal coverage against region size

Marginal coverage is nearly identical across the 15 scores, so it cannot be used to choose among them; the volume ratio to the oracle region separates them.

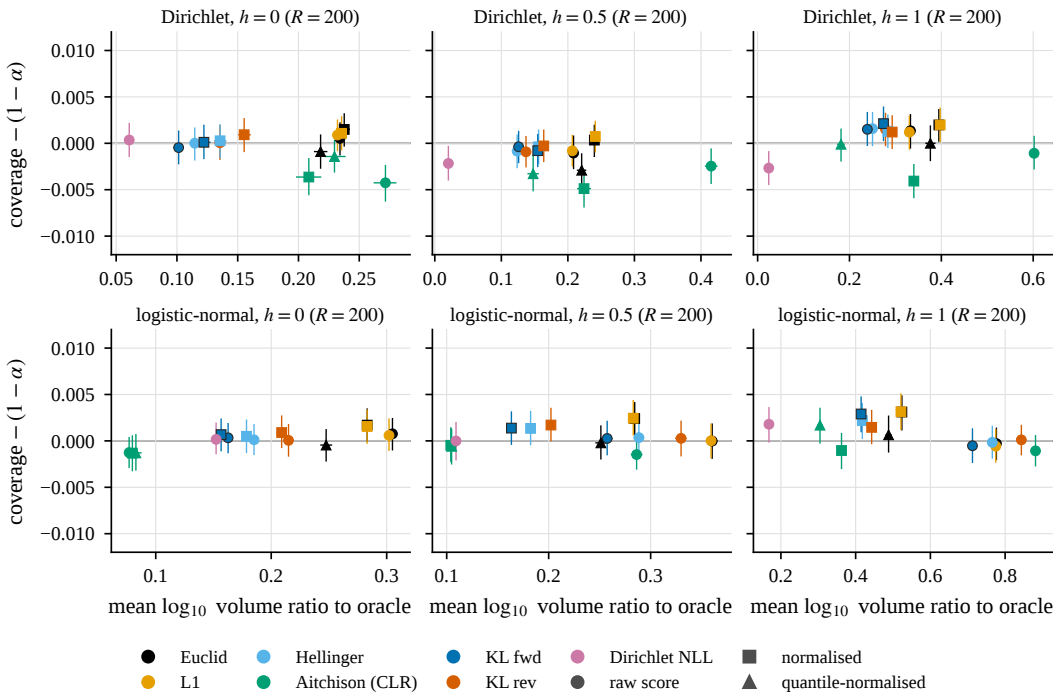


Figure 5: Marginal coverage cannot separate the scores, region size can. True marginal coverage minus 0.9 (ground-truth grid; whiskers: 95% interval over replications) against the mean $\log_{10}$ volume ratio to the oracle region, for 15 scores, $K = 3$, $h = 0, 0.5, 1$. Source: `figures/fig_c3_cov_vs_size.py`.

A.3 Conditional coverage and volume against heteroscedasticity

The full-width version of the conditional-coverage panels of the main text, with the volume ratio against heteroscedasticity.

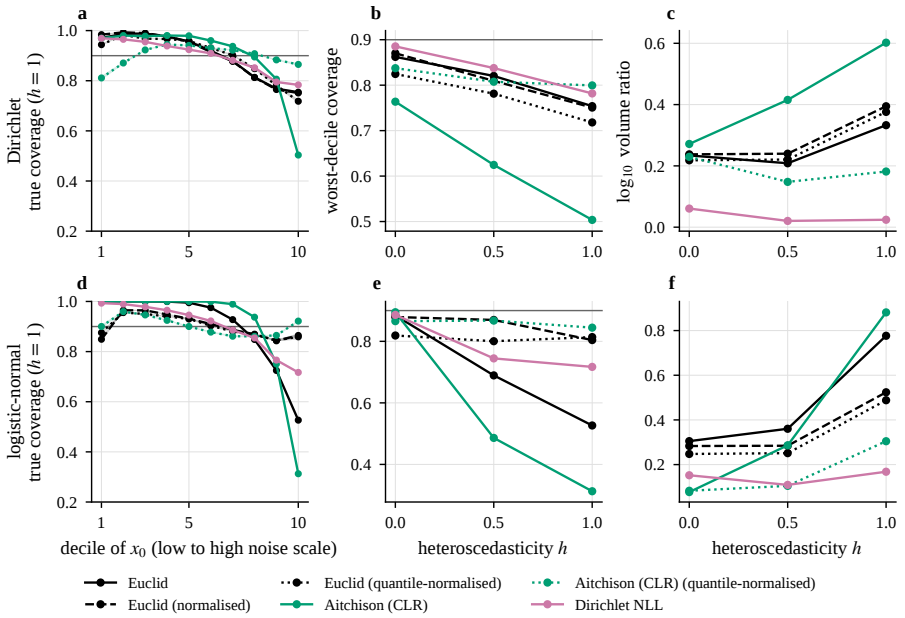


Figure 6: Conditional coverage and region size for six of the 15 scores. Rows: Dirichlet and logistic-normal noise ($K = 3$, $R = 200$). (a,d) True coverage in the ten deciles of x_0 (low to high noise scale) at $h = 1$. (b,e) Worst-decile true coverage and (c,f) $\log_{10}$ volume ratio to the oracle region against the heteroscedasticity h. Source: `figures/fig_c4_conditional.py`.

A.4 Zero replacement

Replacing zeros by δ before the log-ratio transform changes the geometry of boundary strata; this checks how much the conclusions depend on δ.

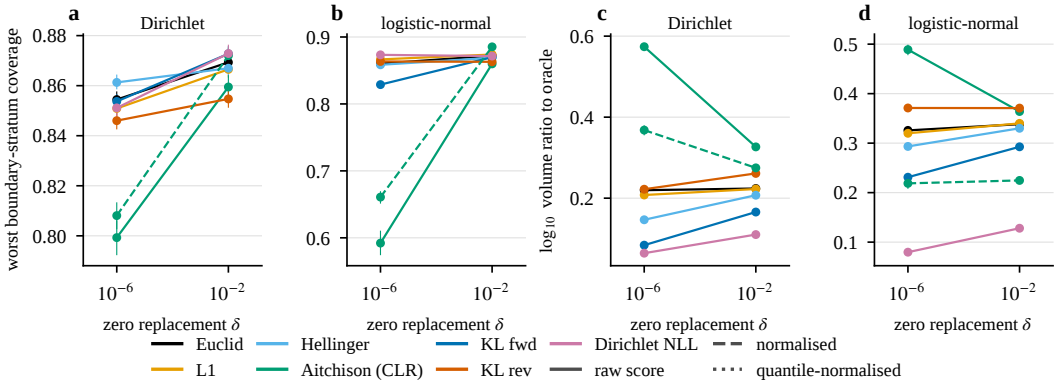


Figure 7: Sensitivity to the zero-replacement constant $\delta \in \{10^{-6}, 10^{-2}\}$ on rounded data, one line per score: (a,b) worst boundary-stratum coverage and (c,d) $\log_{10}$ volume ratio, for Dirichlet and logistic-normal noise (`A1d_summary.csv`). Source: `figures/fig_a1d_delta.py`.

A.5 Bias of the worst-cell estimate

Supports Proposition 2 and the bias-by-score statement in Section 4.

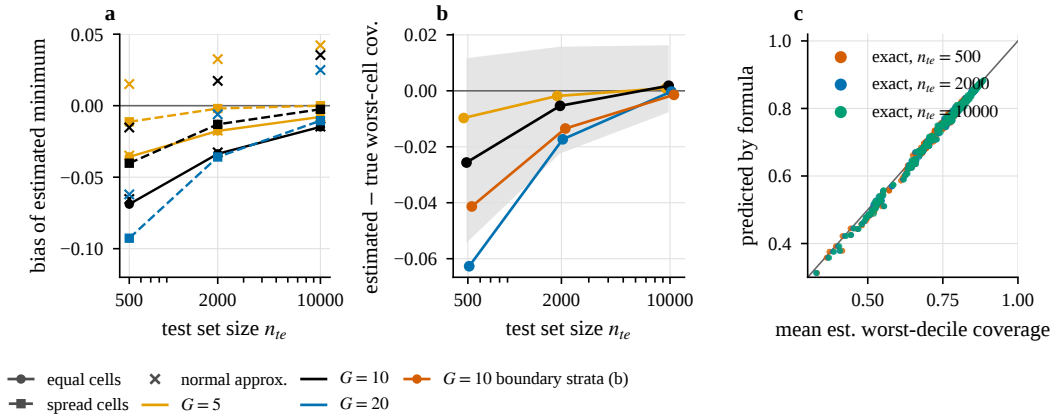


Figure 8: The worst-cell estimate is biased downward by an amount set by the cell size, and Proposition 2 gives it. (a) Bias of the estimated minimum of G cell coverages against the test size, for equal cells and cells with spread coverages (mean coverage 0.9): Monte Carlo (dots) and normal approximation (crosses). (b) Estimated minus true worst-cell coverage in the A1 grid, mean over 20 configurations and 15 scores; grey: 10–90% range for $G = 10$. (c) Exact formula evaluated at each replication's true cell coverages against the mean estimated worst-decile coverage. Source: `figures/fig_c6_worstcell_bias.py`.

A.6 Best score per configuration

The full list of winners that the main text summarises in words.

Table 3: Best score per configuration of A1 by region size (smallest $\log_{10}$ volume ratio), worst-decile coverage and worst boundary-stratum coverage, with the value in brackets; "= name" marks a runner-up whose paired 95% interval for the difference contains zero. R: replications. Volume is not computed for $K = 10$. Source: `tables/tab_c5_regimes.py`.

config.	R	smallest volume	best worst decile	best worst boundary
Dirichlet (base precision 20)				
$h = 0$	200	DirNLL (0.06)	DirNLL (0.89)	DirNLL (0.87)
base	200	DirNLL (0.02)	DirNLL (0.84)	DirNLL (0.87) = Euc/n
prec. 5	100	Ait/nq (0.04) = DirNLL	DirNLL (0.85) = Ait/nq	DirNLL (0.87) = Hel/n
prec. 5, round 2	100	DirNLL (0.07)	Ait/nq (0.86)	Hel/n (0.87)
prec. 80	100	DirNLL (0.03)	L1/n (0.85)	DirNLL (0.88)
quadratic scale	200	DirNLL (0.17)	Ait/nq (0.83)	DirNLL (0.87)
$h = 1$	200	DirNLL (0.02)	Ait/nq (0.80)	Euc/n (0.86)
$K = 5, h = 0$	100	KLf (0.05) = Hel	KLr (0.88) = DirNLL	KLr (0.89)
$K = 5$	100	DirNLL (0.12)	L1/n (0.86) = KLr/n	KLr/n (0.89) = L1
$K = 10$	100	n/a	Euc/nq (0.84)	KLr/n (0.85) = KLr
logistic-normal (base $\sigma = 0.5$)				
$h = 0$	200	Ait (0.08)	Ait (0.89)	Ait (0.89)
base	200	Ait/n (0.10) = Ait/nq	KLr/n (0.88)	Ait/n (0.89)
quadratic scale	200	Ait/nq (0.15)	Ait/nq (0.85)	Ait/nq (0.88)
$\sigma = 0.25$	100	Ait/n (0.06)	KLf/n (0.88) = Hel/n	Ait/n (0.89)
$\sigma = 1$	100	DirNLL (0.10)	KLr/n (0.88)	Ait/n (0.89) = KLf/n
$\sigma = 1$, round 2	100	DirNLL (0.09)	KLr/n (0.88) = Hel/n	Hel/n (0.88)
$h = 1$	200	DirNLL (0.17)	Ait/nq (0.84)	Ait/nq (0.88) = KLf/n
$K = 5, h = 0$	100	Ait (0.15) = Ait/n	Ait (0.89)	Ait (0.89)
$K = 5$	100	Ait/nq (0.18) = Ait/n	KLf/n (0.88)	L1/n (0.89) = KLf/n
$K = 10$	100	n/a	KLf/n (0.88)	KLf/n (0.89) = Hel/n

A.7 Rounding and ties

Rounded data produce ties among scores; breaking ties at random keeps exact coverage.

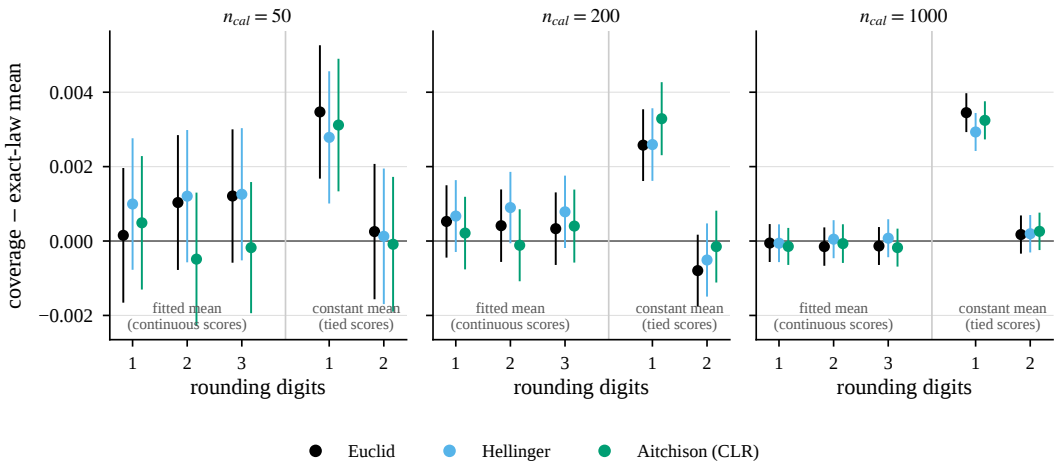


Figure 9: Rounded and tied scores keep the coverage law when ties are broken at random. Observed mean coverage minus the exact-law mean against the calibration size, with 95% intervals, for rounded observations (fitted mean) and tied scores (intercept-only mean). Source: `figures/fig_c2_rounding.py`.

A.8 Covariate shift

Weighted conformal prediction with a known density ratio, for a covariate shift of x_0.

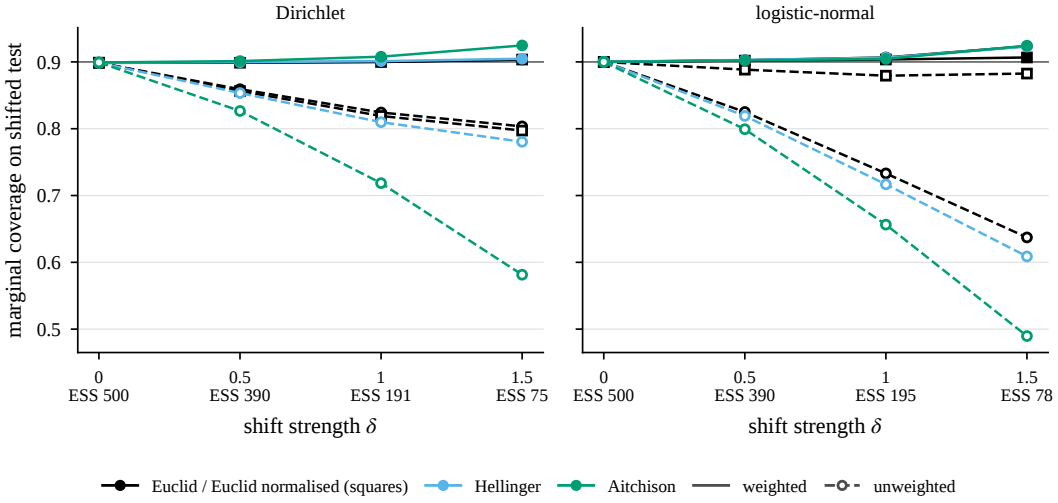


Figure 10: Weighted split conformal prediction under covariate shift of x_0 with known density ratio: coverage against the shift δ (effective sample size in the tick labels), unweighted and weighted. Source: `figures/fig_c11_shift.py`.

A.9 Reversal probability of the worst-cell estimate against the true gap

Probability that the naive worst-decile estimate orders two scores differently from the truth, for the 105 score pairs of 8 configurations with 1000 replications, in bins of the true gap (Section 4.3).

Table 4: Probability that the estimated worst x_0-decile coverage orders two scores differently from the true order, against the true gap, with 95% intervals (bootstrap over configurations), for one replication and for the mean of 25 replications at $n_{\mathrm{te}} = 500, 2000, 10^4$. Last row: smallest gap from which the probability is at most 0.05. Source: `A3_flip_bins.csv`, `A3_flip_threshold.csv`; `tables/tab_c15_flip.py`.

	single replication			mean of 25 replications		
true gap / n_{te}	500	2000	10000	500	2000	10000
0.00–0.005	.48 (.48–.48)	.45 (.45–.46)	.40 (.39–.42)	.41 (.38–.45)	.35 (.32–.39)	.26 (.23–.30)
0.005–0.01	.44 (.43–.45)	.37 (.35–.38)	.25 (.23–.27)	.28 (.24–.31)	.14 (.11–.18)	.07 (.03–.12)
0.01–0.02	.39 (.38–.40)	.28 (.27–.29)	.13 (.11–.16)	.18 (.13–.21)	.05 (.03–.07)	.02 (<.01–.04)
0.02–0.03	.34 (.32–.35)	.19 (.17–.20)	.06 (.05–.08)	.09 (.05–.13)	.02 (<.01–.03)	.01 (<.01–.02)
0.03–0.05	.26 (.24–.28)	.10 (.09–.12)	.02 (.01–.03)	.03 (.01–.06)	<.01	<.01
0.05–0.10	.16 (.14–.18)	.04 (.03–.04)	<.01	<.01	0	0
0.10–0.20	.05 (.04–.06)	<.01	0	0	0	0
≥0.20	<.01	<.01	0	0	0	0
gap with flips ≤ 5% from	0.2	0.05	0.03	0.03	0.01	0.01

A.10 Plug-in version of Proposition 2

Proposition 2 evaluated at cell coverages estimated on the same test set, against the true cell coverages (oracle) and a shrunk variant, for $G = 10$ cells, 8 configurations, 15 scores and 1000 replications.

Table 5: Observed bias of the naive worst-cell estimate, bias predicted by the formula at the true cell coverages (oracle), at estimated ones (plug-in) and at shrunk ones, RMSE of the naive estimate and of the estimate corrected by the plug-in and shrunk bias, correlation r across scores between the observed and the plug-in bias, and single-replication Kendall τ with the true ranking for the naive and the plug-in-corrected estimate. Source: `A3pi_summary.csv`, `A3pi_rank.csv`; `tables/tab_c13_plugin.py`.

		bias of the worst-cell estimate				RMSE			corr.	Kendall τ	
strata	n_{te}	obs.	oracle	plug-in	shrunk	naive	plug-in	shrunk	r	naive	plug-in
x_0 deciles	500	−0.018	−0.022	−0.018	−0.029	0.060	0.060	0.061	0.94	0.62	0.60
	2000	−0.002	−0.007	−0.006	−0.009	0.031	0.031	0.032	0.86	0.71	0.71
	10000	+0.003	−0.001	−0.001	−0.002	0.017	0.017	0.017	0.64	0.76	0.76
boundary strata	500	−0.040	−0.041	−0.027	−0.046	0.061	0.052	0.052	0.94	0.35	0.33
	2000	−0.013	−0.015	−0.011	−0.016	0.027	0.026	0.027	0.88	0.48	0.46
	10000	−0.001	−0.004	−0.003	−0.005	0.013	0.013	0.014	0.64	0.56	0.56

A.11 CV+ and the jackknife+ against split calibration

Paired differences to split calibration on the same draws, Dirichlet and logistic-normal noise with four scores (8 cells), 100 replications; the jackknife+ was not run at $n = 200$.

Table 6: CV+ and jackknife+ minus split: median and range over the 8 cells, and the number of cells whose paired 95% interval lies below or above zero, for marginal coverage, worst x_0-decile coverage and $\log_{10}$ volume ratio. Source: `A5_paired_vs_split.csv`; `tables/tab_c12_cvplus.py`.

n	method	median	range over 8 cells	CI < 0	CI > 0
marginal coverage, difference to split					
30	CV+	-0.015	-0.030 to -0.006	4	0
	jackknife+	-0.035	-0.047 to -0.031	8	0
50	CV+	0.000	-0.019 to $+0.013$	2	1
	jackknife+	-0.017	-0.024 to -0.007	7	0
100	CV+	$+0.009$	-0.001 to $+0.021$	0	5
	jackknife+	-0.002	-0.006 to $+0.008$	0	0
200	CV+	$+0.007$	-0.003 to $+0.010$	0	5
worst x_0-decile coverage, difference to split					
30	CV+	-0.042	-0.108 to $+0.022$	4	0
	jackknife+	-0.084	-0.131 to -0.026	7	0
50	CV+	-0.015	-0.073 to $+0.044$	4	2
	jackknife+	-0.040	-0.088 to $+0.009$	7	0
100	CV+	$+0.011$	$+0.001$ to $+0.052$	0	4
	jackknife+	-0.005	-0.010 to $+0.035$	0	2
200	CV+	$+0.018$	-0.001 to $+0.030$	0	6
$\log_{10}$ volume ratio, difference to split					
30	CV+	-0.163	-0.173 to -0.126	8	0
	jackknife+	-0.197	-0.266 to -0.167	8	0
50	CV+	-0.080	-0.122 to -0.043	8	0
	jackknife+	-0.109	-0.141 to -0.099	8	0
100	CV+	-0.026	-0.042 to -0.018	5	0
	jackknife+	-0.049	-0.069 to -0.026	8	0
200	CV+	-0.015	-0.032 to -0.007	4	0

A.12 Misspecified mean

The true logit mean has a quadratic part ($n_l = 0.5, 1$: 30% and 63% of its variance is not explained by the linear fit); the fitted mean is linear in clr coordinates ($K = 3$, $h = 0.5$, 200 replications).

Table 7: Misspecified mean. Range over the 15 scores of the true marginal coverage and of the increase in $\log_{10}$ volume ratio against correct specification, Kendall τ between the ranking of the scores under misspecification and under correct specification (by volume, worst x_0 decile, worst boundary stratum), and the best score by volume. Source: `A6_summary.csv`, `A6_vs_A1.csv`, `A6_rank_agreement.csv`; `tables/tab_c14_misspec.py`.

		vs correct specification		Kendall τ of the ranking by			best score
noise	n_l	true marg. cov.	$\Delta \log_{10}$ vol.	volume	worst decile	worst bdry.	by volume
Dirichlet	0.5	0.895–0.900	0.31–0.54	+0.39	+0.20	+0.81	DirNLL → Euc/nq
	1	0.891–0.901	0.50–0.79	+0.07	+0.07	+0.60	DirNLL → KLf/n
logistic-normal	0.5	0.896–0.901	0.34–0.65	+0.18	+0.52	-0.10	Ait/n → DirNLL
	1	0.895–0.901	0.45–0.81	+0.18	-0.12	+0.14	Ait/n → DirNLL

B Reproducibility, compute and provenance

All numbers in the paper are read from `results/tables/*.csv`, which the scripts in `code/` regenerate from the raw replications in `results/raw/` (gzip CSV; every file records the configuration, the seeds and the revision of the code that made it). `results/CLAIMS_MAP.md` maps each claim to its file, number and uncertainty; `figures/MAP.md` maps each figure and table to its generator and input files. The step-by-step commands are in `REPRODUCE.md`; we summarise them here.

Environment and data. Python 3.11.2, NumPy 2.4.6, SciPy 1.17.1, pandas 3.0.6, PyTorch 2.14.1 (CPU; used for the Dirichlet-regression likelihood and the CIFAR-10 CNN); no GPU. Synthetic data are generated by `code/cpsimplex/dgp.py` with fixed population coefficients (`coef_seed=12345`). CIFAR-10 [Krizhevsky, 2009] was obtained as the parquet copy of the Hugging Face dataset `uoft-cs/cifar10` (50 000 training and 10 000 test images; md5 of the files used: train `7170dcda...`, test `a52165db...`); the official archive (md5 `c58f3010...`) was not used because the download stalled, and we did not check the two copies against each other by pixel checksum. The creators state no licence. The CNN is a small convolutional network trained for 15 epochs on 40 000 training images; the logits of the 20 000 held-out images are committed (`results/cifar_logits.npz`), so the conformal experiments do not depend on retraining.

Seeds. Replication i of an experiment uses seed $\text{seed}_0 + i$ with $\text{seed}_0 = 1000$ (A1), 50000 (A2), 7000 (A4), 9000 (A5) and 0 (B); the B-synth pool of replication i uses seed $10000+i$; the ground-truth grid uses seed 777.

Compute. CPU only, 2 CPUs per job, on a shared host that was heavily loaded, so the wall times below are upper bounds: one A2 file 15–40 min (45 files); one A1 configuration 0.5–6 h (volume computation dominates; $K = 10$ slowest); A3 half-split 0.5–2 h per configuration (8); A4 about 1 h; B-synth about 1 h; CNN training about 4 h; B-cifar about 5 min per configuration; analysis scripts a few minutes each (`analyze_a3.py` about 10 min).

Discarded and failed runs. A first A1 grid used 30 evaluation points per slab and was discarded because the ground-truth conditional coverage was too noisy; it was rebuilt with 300 points and 30 draws per slab. A first B-synth run estimated size-stratified coverage without a minimum cell size (errors of 0.1–0.2 from tiny cells) and was discarded; the rerun applies the minimum-30-point rule. The first A2 files did not record n_{te}, α or the code revision and were replaced by a rerun with explicit settings. A first DGP with extreme means (coefficient scale 0.8) had an oracle region with zero Monte-Carlo volume and was replaced by scale 0.4. Jobs of the half-split study were reported as failed by the scheduler because the launcher script was edited while they ran; their output files are complete (1000 replications each). No number from a discarded run is used. Not done: real compositional data, volumes for $K = 10$, misspecifications other than the one quadratic mean, and jackknife+ at $n = 200$.

Literature restriction. The task excludes one published paper on its own question, and any page citing or discussing it. During a broad literature search for conformal prediction with multivariate and Dirichlet-related topics on 2026-11-04, a search listing appeared whose title is a conformal-prediction paper on compositional data with Dirichlet regression; only its title and about 600 characters of its abstract snippet were seen, the page was not opened, and nothing from it was used. We cannot tell whether it is the excluded paper or a paper discussing it, and treat it as excluded; we did not search for it or for its references. Later searches by the authors on adaptive prediction sets, conditional-coverage limits and multivariate-response conformal prediction on the same day surfaced nothing on compositional or Dirichlet conformal prediction. The related work and its statements of novelty are therefore relative to the sources we consulted.

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